



NORTH SYDNEY BOYS HIGH SCHOOL

2011 YEAR 12 HSC ASSESSMENT TASK 1

Mathematics

Extension 2

Examiner: S. Ireland

General Instructions

- Working time – **55 minutes**
- Write on both sides of the paper (with lines) in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Fletcher
- ☐ Mr Ireland
- ☐ Mr Rezcallah

Student Number:

Question	1	2	3	4	5	6	Total	Total
Mark	$\overline{11}$	$\overline{9}$	$\overline{10}$	$\overline{8}$	$\overline{7}$	$\overline{8}$	$\overline{53}$	$\overline{100}$

Question 1 (11 marks) Start a new page.**Marks**

- (a) Let z be the complex number $2 \operatorname{cis}(-\frac{\pi}{3})$. On a single, half-page Argand diagram, plot the points: z , $\bar{z}$, z^2 , and $\frac{1}{z}$. 4
- (b) Given $z = 1 - i$ and $\omega = -1 + i\sqrt{3}$,
- (i) Express $\frac{1}{z}$ in $a + ib$ form (where a and b are real) 1
 - (ii) Calculate ωz in $a + ib$ form 1
 - (iii) Find $\arg z$ and $\arg \omega$ 2
 - (iv) Hence find $\arg(\omega z)$ 1
 - (v) Hence prove $\sin \frac{5\pi}{12} = \frac{\sqrt{2}(1+\sqrt{3})}{4}$ 2

Question 2 (9 marks) Start a new page.

- (a) If ω is a non-real root of $z^3 = 1$, then
- (i) Show that $1 + \omega + \omega^2 = 0$ 1
 - (ii) Evaluate $(1 - 3\omega + \omega^2)(1 + \omega - 8\omega^2)$ 2
- (b) If $1 + i$ is a zero of $P(x) = 3x^3 - 7x^2 + 8x - 2$, then
- (i) Find all zeroes of $P(x)$ 2
 - (ii) Factorise $P(x)$ over $\mathbb{R}$ 1
- (c) Factorise $P(x) = 3x^4 + 8x^3 + 6x^2 - 1$ completely given that $P(x) = 0$ has a root of multiplicity 3. 3

Question 3 (10 marks) Start a new page.

- (a) (i) Find real numbers x and y such that $(x + iy)^2 = -3 - 4i$ 2
- (ii) Hence solve the equation $z^2 - 3z + (3 + i) = 0$ 2
- (b) On separate Argand diagrams sketch the locus of a point which satisfies:
- (i) $|z - 1| = |z - i|$ 2
 - (ii) $z^2 - (\bar{z})^2 = 16i$ 2
- (c) Sketch the region satisfying both $0 \leq \arg(z - 2) \leq \frac{\pi}{2}$ and $\operatorname{Im}(z) \leq 1$ 2

Question 4 (8 marks) Start a new page.

Given that $x^3 - 2x^2 - 5 = 0$ has roots α, β, γ ,

(a) Evaluate $\alpha^2 + \beta^2 + \gamma^2$ 2

(b) Evaluate $\alpha^3 + \beta^3 + \gamma^3$ 2

(c) Write equations with roots:

(i) $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ 2

(ii) $\alpha + \beta + 2\gamma, \alpha + 2\beta + \gamma, 2\alpha + \beta + \gamma$ 2

Question 5 (7 marks) Start a new page.

(a) Sketch z_1 and z_2 on an Argand diagram given that $|z_1| = |z_2| = |z_1 - z_2|$
Hence find the magnitude of $\arg\left(\frac{z_1}{z_2}\right)$ 2

(b) (i) Sketch $|z - 15i| \leq 5$ on an Argand diagram. 2

(ii) Find the complex number z with the least positive argument satisfying $|z - 15i| \leq 5$. (Give your answer in $a + ib$ form). 3

Question 6 (8 marks) Start a new page.

(a) (i) Show the roots of $z^5 = 1$ on a unit circle on an Argand diagram. 3

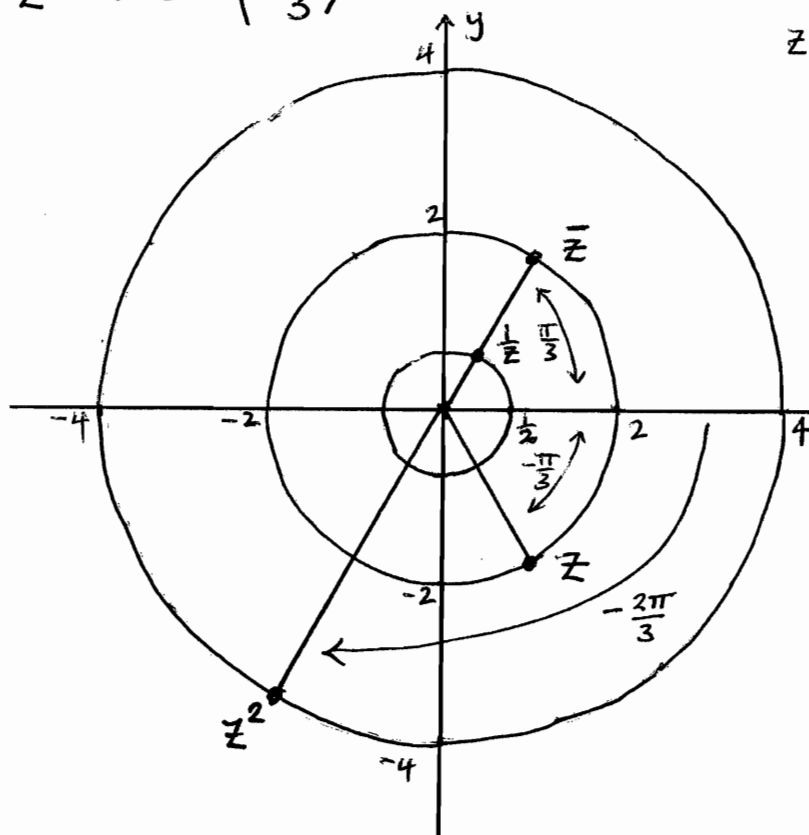
(ii) Hence show that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$ 2

(b) Sketch the locus $\arg\left(1 - \frac{1}{z}\right) = \frac{\pi}{4}$ and find its Cartesian equation. 3

END OF EXAMINATION

Q1

(a) $z = 2 \operatorname{cis}\left(-\frac{\pi}{3}\right)$



$$\bar{z} = 2 \operatorname{cis} \frac{\pi}{3}$$

$$z^2 = 4 \operatorname{cis} -\frac{2\pi}{3}$$

$$\frac{1}{z} = \frac{1}{2} \operatorname{cis} \frac{\pi}{3}$$

[ALT: $z = 1 - \sqrt{3}i$
 $z^2 = -2 - 2\sqrt{3}i$
 $\bar{z} = -2 + 2\sqrt{3}i$
 $\frac{1}{z} = \frac{1}{4} + \frac{\sqrt{3}}{4}i$]

(* $\frac{1}{z}$ & $\bar{z}$ must have same arg.
 * must show data on sketch.)

✓✓✓✓

(b) (i) $\frac{1}{z} = \frac{1}{2} + \frac{1}{2}i$ ✓

(ii) $wz = (\sqrt{3} - 1) + i(\sqrt{3} + 1)$ ✓

(iii) $z = 1 - i \quad \therefore \arg z = -\frac{\pi}{4}$ ✓

$w = -1 + i\sqrt{3} \quad \therefore \arg w = \frac{2\pi}{3}$ ✓

(iv) $\arg(wz) = \frac{5\pi}{12}$ ✓

(v) $\sin \frac{5\pi}{12} = \frac{\operatorname{Im}(wz)}{|wz|}$

$= \frac{\operatorname{Im}(wz)}{|w| \cdot |z|}$

$= \frac{\sqrt{3} + 1}{2 \cdot \sqrt{2}}$

[* must use previous part of question]

[ALT: $wz = 2\sqrt{2} \operatorname{cis} \frac{5\pi}{12}$
 & $wz = (\sqrt{3} - 1) + i(\sqrt{3} + 1)$, then equate imag. parts]

$\therefore \sin \frac{5\pi}{12} = \frac{\sqrt{2}(\sqrt{3} + 1)}{4}$
 as required. ✓✓

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Question 2:

(a) (i) $w^3 = 1$

$$w^3 - 1 = 0$$

$$(w-1)(w^2 + w + 1) = 0.$$

$$\therefore w^2 + w + 1 = 0$$

(ii) $(1-3w+w^2)(1+w-8w^2)$

$$= (1+w^2-3w)(1+w-8w^2)$$

$$= (-w-3w)(-w^2-8w^2) = (-4w)(-9w^2)$$

$$= +(4w)(9w^2) = 36w^3 = 36.$$

(b) (i) $1+i$ is a zero $\Rightarrow 1-i$ is a zero.

$$S = -\frac{b}{a} = \frac{7}{3}.$$

$$1+i+1-i+\alpha = \frac{7}{3}.$$

$$2 + \alpha = \frac{7}{3}.$$

$$\alpha = \frac{7}{3} - 2 = \frac{1}{3}.$$

(ii) $P(x) = (3x-1)(x^2 - \text{sum } x + P)$

$$\left. \begin{array}{l} \text{Product} = p = (1+i)(1-i) = 1-i^2 = 2. \\ \text{Sum} = S = 1+i+1-i = 2. \end{array} \right\} x^2 - 2x + 2.$$

$$P(x) = (3x-1)(x^2 - 2x + 2).$$

or:
$$\begin{array}{r} \frac{1}{3} \sqrt{\begin{array}{rrrr} 3 & -7 & 8 & -2 \\ & 1 & -2 & 2 \\ \hline 3 & -6 & 6 & 0 \end{array}} \end{array}$$

$$P(x) = \left(x - \frac{1}{3}\right)(3x^2 - 6x + 6)$$

$$= (3x-1)(x^2 - 2x + 2).$$

(c) $P(x) = 3x^4 + 8x^3 + 6x^2 - 1.$

$$P'(x) = 12x^3 + 24x^2 + 12x.$$

$$P''(x) = 36x^2 + 48x + 12 = 12(3x^2 + 4x + 1)$$

$$= 12(3x+1)(x+1)$$

✓ for $(3x-1)(x^2-2x+2)$

N.B:
 $\left(x - \frac{1}{3}\right)(x^2 - 2x + 2)$
won't get the mark.

✓

Question 2 - (cont'd)

$$P'(-1) = P''(-1) = 0.$$

$\therefore x = -1$ is a triple root. ✓

$$S = -\frac{b}{a} = -\frac{8}{3}.$$

$$-1 + -1 + -1 + \beta = -\frac{8}{3}.$$

$$-3 + \beta = -\frac{8}{3}.$$

$$\beta = 3 - \frac{8}{3} = \frac{1}{3}.$$

✓ for $(x+1)^3$

$$\therefore P(x) = (x+1)^3 (3x-1).$$

✓ for correct factors

2nd Method:

$$\begin{array}{r} -1 \overline{) 3 \quad 8 \quad 6 \quad 0 \quad -1} \\ \underline{-3 \quad -5 \quad 1 \quad 1} \\ 3 \quad 5 \quad 1 \quad 1 \quad 0 \end{array}$$

$$(3x^3 + 5x^2 + x + 1)(x+1)$$

$$\begin{array}{r} -1 \overline{) 3 \quad 5 \quad 1 \quad 1} \\ \underline{-3 \quad -2 \quad -1} \\ 3 \quad 2 \quad -1 \quad 0 \end{array}$$

$$(x+1)^2 (3x^2 + 2x - 1)$$

$$P(x) = (x+1)^2 (x+1)(3x-1)$$

$$P(x) = (x+1)^3 (3x-1).$$

Again $P(x) = (x+1)^3 \left(x - \frac{1}{3}\right)$ will not get the mark.

If students do the same mistake twice,
they are penalised once for this error.

Q3

$$(a) (i) (x+iy)^2 = x^2 - y^2 + 2xyi$$

$$\therefore \begin{cases} x^2 - y^2 = -3 \\ xy = -2 \end{cases}$$

$$x^2 - \frac{4}{x^2} = -3$$

$$(x^2 - 1)(x^2 + 4) = 0$$

$$\therefore x = 1 \text{ or } -1 \quad (x \text{ is real})$$

$$\therefore y = -2 \text{ or } 2$$

So answer is $x=1, y=-2$ or $x=-1, y=2$.

$$(ii) z^2 - 3z + (3+i) = 0$$

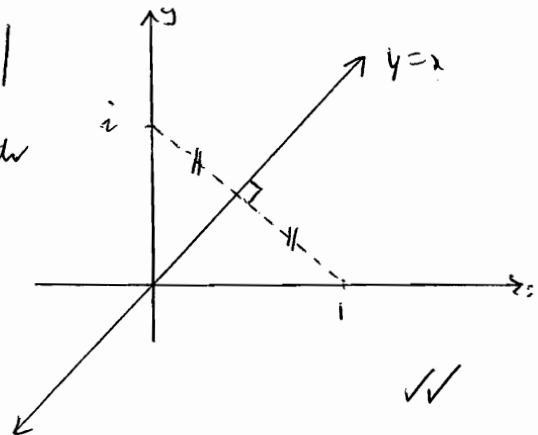
$$z = \frac{3 \pm \sqrt{9 - 4(3+i)}}{2}$$

$$= \frac{3 \pm (1-2i)}{2}$$

$$\therefore z = 2-i \text{ or } 1+i$$

$$(b) (i) |z-1| = |z-i|$$

(* must show perpendicular bisector
or give equation)

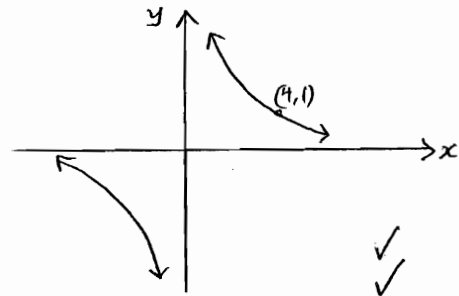


$$(ii) z^2 - (\bar{z})^2 = 16i$$

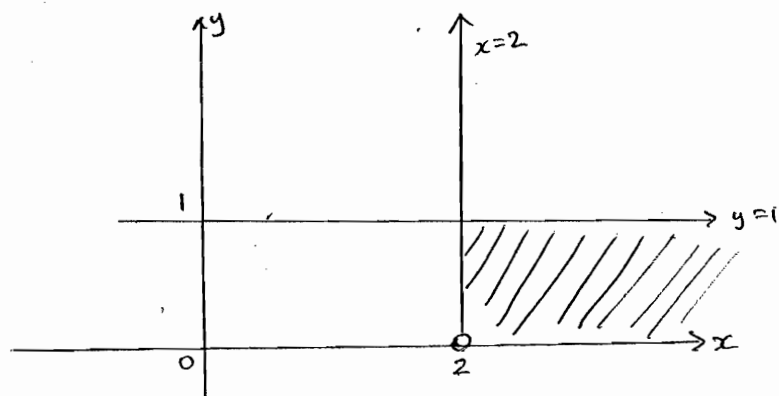
$$\therefore (z - \bar{z})(z + \bar{z}) = 16i$$

$$2yi \cdot 2x = 16i$$

$$xy = 4$$



$$(c) 0 \leq \arg(z-2) \leq \frac{\pi}{2} \text{ and } \operatorname{Im}(z) \leq 1$$



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Q4

$$x^3 - 2x^2 - 5 = 0$$

(a) $\alpha + \beta + \gamma = 2$

$$\alpha\beta + \alpha\gamma + \beta\gamma = 0$$

$$\begin{aligned}\therefore \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \quad \checkmark \\ &= 2^2 - 2(0) \\ &= 4 \quad \checkmark\end{aligned}$$

(b) Sub. α, β, γ into equation :-

$$\begin{aligned}\alpha^3 - 2\alpha^2 - 5 &= 0 \\ + \quad \beta^3 - 2\beta^2 - 5 &= 0 \\ \gamma^3 - 2\gamma^2 - 5 &= 0\end{aligned}$$

$$\alpha^3 + \beta^3 + \gamma^3 - 2(\alpha^2 + \beta^2 + \gamma^2) - 15 = 0 \quad \checkmark$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = 15 + 2(4)$$

$$= 23 \quad \checkmark$$

(c) (i) $\left(\frac{1}{x}\right)^3 - 2\left(\frac{1}{x}\right)^2 - 5 = 0 \quad \checkmark$

$$\frac{1}{x^3} - \frac{2}{x^2} - 5 = 0$$

$$\therefore 1 - 2x - 5x^3 = 0 \quad \checkmark$$

(ie $5x^3 + 2x - 1 = 0$)

(ii) Since $\alpha + \beta + \gamma = 2$, we want roots $\alpha+2, \beta+2, \gamma+2$. $\checkmark$

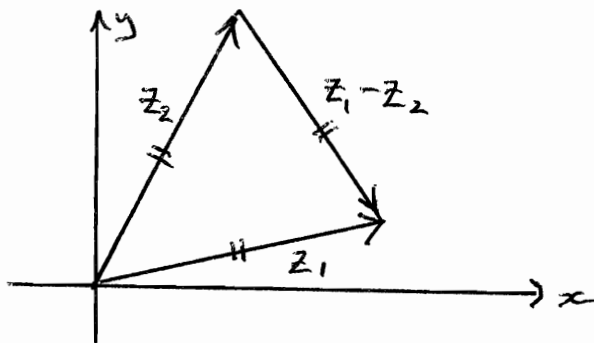
$$\therefore \text{equation is } (x-2)^3 - 2(x-2)^2 - 5 = 0 \quad \checkmark$$

$$\therefore x^3 - 6x^2 + 12x - 8 - 2x^2 + 8x - 8 - 5 = 0$$

$$\therefore x^3 - 8x^2 + 20x - 21 = 0$$

Q5.

(a)



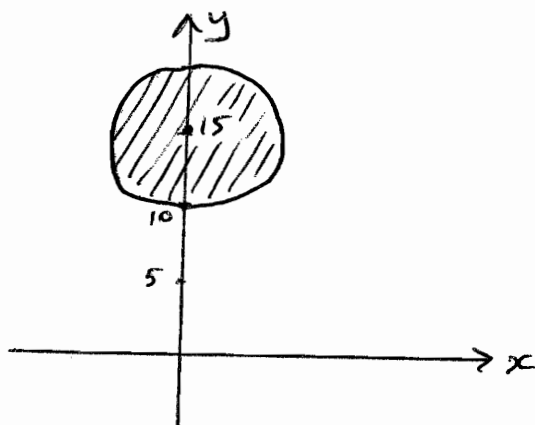
$$\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$$

$$= \pm \frac{\pi}{3} \quad (\text{as } \Delta \text{ is equilateral})$$

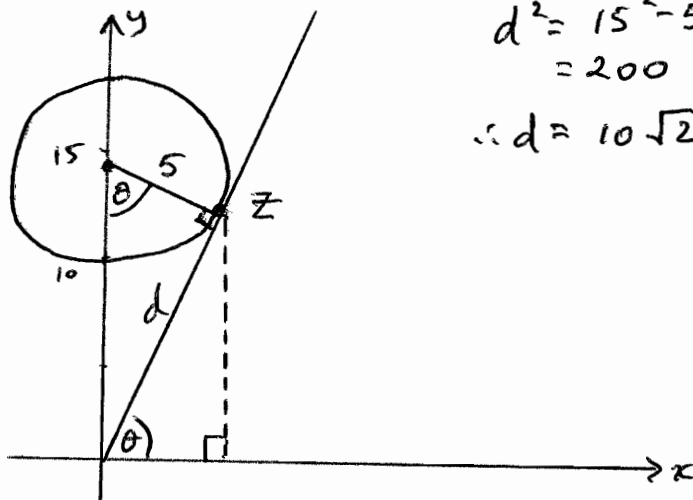
(ie magnitude = $\frac{\pi}{3}$)

(b)

(i)



(ii)



$$\left. \begin{aligned} d^2 &= 15^2 - 5^2 \\ &= 200 \\ \therefore d &= 10\sqrt{2} \end{aligned} \right\} \therefore z \text{ has}$$

$$\begin{aligned} \text{x coordinate} &= d \cos \theta \\ &= 10\sqrt{2} \cdot \frac{5}{15} \\ &= \frac{10\sqrt{2}}{3} \end{aligned}$$

Likewise, z's y coordinate

$$= 10\sqrt{2} \cdot \frac{10\sqrt{2}}{15}$$

$$= \frac{40}{3}$$

$$\therefore z = \frac{10\sqrt{2}}{3} + \frac{40}{3}i$$

✓✓✓

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Q6 - continued

(5-b) 2nd Approach:

$$\operatorname{Re}\left(1 - \frac{1}{z}\right) = \operatorname{Im}\left(1 - \frac{1}{z}\right) \text{ since angle is } \pi/4.$$

$$\text{Let } z = x + iy.$$

$$\frac{1}{z} = \frac{1}{x+iy} = \frac{1}{x+iy} \cdot \frac{x-iy}{x-iy} = \frac{x-iy}{x^2+y^2}.$$

$$1 - \frac{1}{z} = 1 - \frac{(x-iy)}{(x^2+y^2)} = \frac{x^2+y^2-x+iy}{x^2+y^2}.$$

$$\operatorname{Re}\left(1 - \frac{1}{z}\right) = \frac{x^2+y^2-x}{x^2+y^2}.$$

$$\operatorname{Im}\left(1 - \frac{1}{z}\right) = \frac{y}{x^2+y^2}.$$

$$\therefore \frac{x^2+y^2-x}{x^2+y^2} = \frac{y}{x^2+y^2}.$$

$$x^2 - x + y^2 - y = 0.$$

$$\underbrace{x^2 - x + \frac{1}{4}}_{\left(x - \frac{1}{2}\right)^2} + \underbrace{y^2 - y + \frac{1}{4}}_{\left(y - \frac{1}{2}\right)^2} = \frac{1}{4} + \frac{1}{4}.$$

$$\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{2}, \quad y > 0.$$

$\therefore$ Circle of centre $\left(\frac{1}{2}, \frac{1}{2}\right)$ and $r = \frac{1}{\sqrt{2}}.$

Locus is a major arc of this circle.

3rd approach: $\tan^{-1}\left(1 - \frac{1}{z}\right) = \tan^{-1} \pi/4.$

$$\frac{y}{x^2+y^2-x} = 1.$$

$$x^2+y^2-x = y.$$

$$x^2+y^2-x-y=0.$$

$$\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{2}, \text{ where } y > 0.$$

Locus is the major arc of this circle.
(sketched before).

Question 5

(a) $z^5 = 1$.

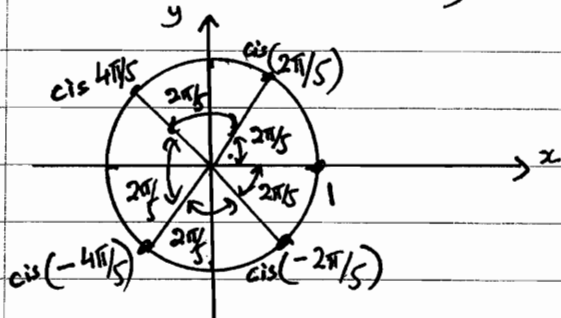
$$(\cos \theta + i \sin \theta)^5 = 1$$

$$\cos 5\theta + i \sin 5\theta = 1.$$

$$5\theta = 2k\pi$$

$$\therefore \theta = \frac{2k\pi}{5} \quad k = 0, \pm 1, \pm 2.$$

Roots are $\text{cis } 0 = 1, \text{cis } \frac{2\pi}{5}, \text{cis } (-\frac{2\pi}{5}), \text{cis } \frac{4\pi}{5}, \text{cis } (-\frac{4\pi}{5})$.



✓✓ for correct circle with the roots shown.

✓ for showing (even in ii) that $\text{cis } 8\pi/5 = \text{cis } (-2\pi/5)$
 $\text{cis } 6\pi/5 = \text{cis } (-4\pi/5)$

(ii) Adding the sum of the roots for $z^5 - 1 = 0$.

$$1 + \text{cis } \frac{2\pi}{5} + \text{cis } (-\frac{2\pi}{5}) + \text{cis } \frac{4\pi}{5} + \text{cis } (-\frac{4\pi}{5}) = 0.$$

From diagram: $1 + \cos 2\pi/5 + \cos(-2\pi/5) + \cos 4\pi/5 + \cos(-4\pi/5) = 0.$ ✓

But $\cos(-2\pi/5) = \cos 2\pi/5$ $\cos(-4\pi/5) = \cos 4\pi/5$.

$$1 + 2 \cos(2\pi/5) + 2 \cos(4\pi/5) = 0.$$

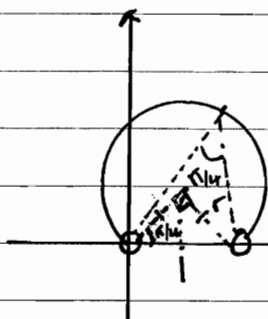
$$2 \cos 2\pi/5 + 2 \cos 4\pi/5 = -1 \quad \div 2.$$
 ✓

$$\cos 2\pi/5 + \cos 4\pi/5 = -\frac{1}{2}.$$

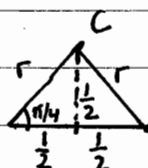
(b) $\arg(1 - \frac{1}{z}) = \pi/4$.

$$\arg\left(\frac{z-1}{z}\right) = \pi/4.$$

$$\cos \pi/4 = \frac{r}{1} \Rightarrow r = \frac{1}{\sqrt{2}}.$$



✓ for correct shape



$$C(\frac{1}{2}, \frac{1}{2}).$$

✓ for centre, radius or equation of circle

Equation of locus: $(x - \frac{1}{2})^2 + (y - \frac{1}{2})^2 = \frac{1}{2}$ where $y \geq 0$.

✓ for correct answers!

N.B: The last mark should show $y \geq 0$ or equivalent in the equation